

EDUCATION

Purdue University, USA
Masters Mechanical Engineering
 GPA: 3.67/4 Aug 2014 – Dec '16
Minor: Computational Science

Vellore Institute of Technology, India
Bachelors Mechanical Engineering
 GPA: 9.03/10 2010 - 2014

SKILLS

Heat Transfer/CFD

- CAD and CFD tools Ansys Icepak, FloTherm XT, COMSOL
- Flow and thermal measurement techniques like IR Camera, Thermocouples

Product Development

- Consumer Electronic Product Development Process
- SolidWorks CAD
- Failure analysis

Programming Languages

- Unix scripting
- MATLAB
- Python

PUBLICATIONS

Passive Thermal Management Using Phase Change Materials: Experimental Evaluation of Thermal Resistances, *ASME Interpack (2015)*
<https://bit.ly/2zMa9eL>

Experimental investigation of Phase Change Materials for electronics cooling, *Intl J of Thermal Sciences (2018)*
<https://bit.ly/2zXyxZb>

Investigation of aluminum foams and graphite fillers for improving the thermal conductivity of paraffin wax-based phase change materials, *ng, IEEE ITherm (2017)*
<https://shorturl.at/AGEkQ>

SUMMARY

- **Thermal Design & Testing:** Deep technical and functional expertise in conduction, convection, radiation, numerical modeling, developing thermal architectures, characterizing thermal performance, developing thermal mitigation strategies
- **Consumer Electronics:** 5+ years experience in fast-paced, high volume consumer electronics (iPads, smart speakers, robots) product development process (xVT) with a solid understanding of requirements at each stage and communicating thermal performance to cross functional teams, product managers and executives

PROFESSIONAL EXPERIENCE

Apple Inc 28 Aug 2023 – Present
System Product Design Engineer (Thermal Engineer ICT 4) Austin, TX, USA

Thermal testing, design of Apple iPad Air (6th/7th gen, 2024/2025), iPhone 17 (2025)

- Ensures all thermal objectives, from initial concept stage through fruition and post launch, are met or exceeded to delight Apple customers by applying expertise in heat transfer (conduction, convection, radiation), thermal measurement techniques and software thermal management
- Conduct Field failure analysis to determine root cause and corrective action
- Develops and implements frameworks to continually optimize thermal testing and data analysis processes

Amazon Lab 126 17 Dec 2018 – 21 July 2023
Thermal Engineer II (Applied Scientist II) Sunnyvale, USA

Thermal testing, design of Amazon's flagship Echo Dot (3rd, 5th Gen), Echo (3rd Gen), Echo Show 10, Astro Robot, Halo Rise Echo Studio and Echo Flex

- Utilized CAD and CFD tools to drive thermal design of new products and collaborate closely with other engineering and product teams
- Enabled \$250k cost savings for Echo Dot by validating & assessing thermal risk during new product development
- Led thermal validation of Amazon first consumer robot (Astro) by utilizing flow and thermal measurement techniques to ensure thermal health and safety
- Developed a novel framework in Python which reduced test time by 50% by automating music/video streaming, control display/LED ring brightness, recording internal temperatures, powers
- Developed Temperature Sensor feature on Echo Dot (5th gen) by collecting extensive device data using Python and Unix scripting to process and visualize large volume of data

Syscon International 20 Feb 2017 – Oct 19 2018
Product Engineer I South Bend, USA

Developed portable temperature measurement systems from concept to manufacturing

- Supported new product development initiatives according to design and market needs
- Performed and standardized calibration using thermocouples
- Performed hardware and software prototype testing
- Made products faster by 25% by developing algorithms to
- Process field data using MATLAB
- Trained and provided field and remote technical support to distributors and sales team
- Presented at tradeshow and conferences of the iron and steel industry (American Foundry Society - AFS and Association for iron and steel technology - AIST)

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Purdue University**Graduate Research Assistant**

1 Jan 2015 – 6 Dec 2016

West Lafayette, USA

- Developed a cooling solution for consumer electronics resulting in 5% reduction in processor temperatures
- Performed thermal design and analysis of consumer electronic devices using cfd analysis tools such as COMSOL, Ansys FLUENT
- Identified and recommend optimal material properties and packaging design to maximize thermal performance and minimize material cost and weight
- Acquired hands-on knowledge of CFD (computational fluid dynamics) through a graduate level course 'Numerical Methods in Heat Transfer' (me608)
- Published and presented master's thesis and research papers to the thermal engineering community at leading conferences (ASME Interpack) and journals (International Journal of Thermal Sciences)
- Developed CAD models of consumer electronic products such as smartphones in Solidworks CAD application from teardowns and literature review
- Analyzed product temperatures and optimized thermal design to minimize temperatures and meet product performance requirements
- Evaluated and validated proposed thermal solution using temperature measurement tools (IR camera, thermocouples)

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